Customer Shopping Behavior Analysis

1. **Project Overview**

   This project studies customer shopping habits using data from 3,900 purchases across different product categories. The main goal is to understand how much customers spend, what products they prefer, how often they shop, and whether they use subscription or discounts. These insights can help businesses understand their customers better and make better business decisions.

2 . Dataset Summary

1. Rows – 3,900
2. Columns – 18
3. Key Features

- Customer demographics : Age, Gender, Location and Subscription Status
- Purchase Details – Item purchased, Category, Purchase Amount, Size, Season , Colour.
- Shopping Behaviour – Discount Applied, Promocode Applied, Frequency of Purchase, Previous Purchases, Review and Shipping details.

3.Exploratory Data Analysis using Python

- Data loading : Imported the data using pandas
- Exploration : Used df.info() to check structure and .describe() for summary statistics.

```
[5]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  3900 non-null   str
 3   Item Purchased          3900 non-null   str
 4   Category                3900 non-null   str
 5   Purchase Amount (USD)   3900 non-null   int64
 6   Location                3900 non-null   str
 7   Size                    3900 non-null   str
 8   Color                   3900 non-null   str
 9   Season                  3900 non-null   str
 10  Review Rating           3900 non-null   float64
 11  Subscription Status     3900 non-null   str
 12  Shipping Type           3900 non-null   str
 13  Discount Applied        3900 non-null   str
 14  Promo Code Used         3900 non-null   str
 15  Previous Purchases      3900 non-null   int64
 16  Payment Method          3900 non-null   str
 17  Frequency of Purchases  3900 non-null   str
dtypes: float64(1), int64(4), str(13)
memory usage: 548.6 KB
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)
count	3900.000000	3900.000000	3900	3900	3900	3900.000000
unique	NaN	NaN	2	25	4	NaN
top	NaN	NaN	Male	Blouse	Clothing	NaN
freq	NaN	NaN	2652	171	1737	NaN
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392
min	1.000000	18.000000	NaN	NaN	NaN	20.000000
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000

- Column Standardization : Used snakecase for better documentation and readability
- Additional Columns : Created **age_group** column into 4 parts using qcut function and created **purchase_frequency_days** column by making use of dictionaries.
- Database Integration : Connected Python script to MySQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using MySQL.

1. What is the total revenue generated by male vs. female customers?

	gender	revenue
▶	Male	157890
	Female	75191

2. High-Spending Discount Users – Identified customers who used discounts but still spent above the average purchase amount.

	customer_id	purchase_amount
▶	2	64
	3	73
	4	90
	7	85
	9	97
	12	68
	13	72
	16	81
	20	90
	22	62
	24	88
	29	94
	32	79
	33	67
	35	91
	37	69
	40	60
	41	76
	43	100
	44	69
	55	94

customer 29 ×

3. Top 5 Products by Rating – Found products with the highest average review ratings.

	item_purchased	Average Product Rating
	Gloves	3.86
	Sandals	3.84
	Boots	3.81
	Hat	3.81
	T-shirt	3.78

4. Shipping Type Comparison – Compared average purchase amounts between Standard and Express shipping.

	shipping_type	ROUND(AVG(purchase_amount),2)
▶	Express	60.48
	Standard	58.46

5. Subscribers vs. Non-Subscribers – Compared average spend and total revenue across subscription status.

subscription_status	total_customers	avg_spend	total_revenue
Yes	1053	59.49	62645
No	2847	59.87	170436

6. Discount-Dependent Products – Identified 5 products with the highest percentage of discounted purchases.

item_purchased	discount_rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

7. Customer Segmentation – Classified customers into New, Returning, and Loyal segments based on purchase history

customer_segment	Number of Customers
Loyal	3116
Returning	701
New	83

8. Top 3 Products per Category – Listed the most purchased products within each category.

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

9. Repeat Buyers & Subscriptions – Checked whether customers with >5 purchases are more likely to subscribe

subscription_status	repeat_buyers
Yes	958
No	2518

10. Revenue by Age Group – Calculated total revenue contribution of each age group

age_group	total_revenue
Young_Adult	62143
Middle_Aged	59197
Adult	55978
Senior	55763

5. Business Recommendations

- Increase Subscriptions : By promoting benefits for subscribers.
- Customer Retention Programs : Reward regular buyers to move them into the “Loyal” segments .

- Product Positioning : Move/Highlight top-rated and best-selling products in campaigns across all customers .
- Targeted Marketing : Focus efforts on high-revenue age groups and express-shipping users.